PROGRESS REPORT

Gloxad Academy
Platform build

Everything built so far across the web app and the backend, what was tested, what remains, and what is needed next.

www.gloxadacademy.com

Prepared 2 November 2026 | Web app: Gloxadapp | Backend: gloxard-backend

THE STATE OF THE PROJECT

1. Summary

Gloxad Academy is an online learning platform made of two code bases: a Next.js web app and an Express and MongoDB backend. Work started on 20 to 22 September 2026. Since then the backend has grown into a full learning platform API, the web app covers sign-up to certificate for learners, tutors and admins, and the trimester LMS (cohort programme) is partly live. This week's assignment, the Trimester Program and LMS landing pages, is under way.

72
commits across both repositories (37 web, 35 backend)

74
free courses in the catalogue

15+
backend build units delivered

2 of 4
LMS landing pages built this week

Where things stand

- **Working and tested:** accounts and sign-in, course catalogue, enrolment and payment plumbing, the learning engine, the redesigned course player, quizzes, certificates with public verification, the learner home, messaging and notifications, and (new) the Trimester and Scholarships pages.
- **Partly built:** the trimester LMS student screens. Dashboard, weekly path, live class, coding lab and help centre use live data. Lesson, quiz, assignments, community, certificates and the instructor and admin dashboards still show sample data.
- **Not built yet:** assignments and grading, the Progress and Final Project screen, discussions, scholarship and programme admin screens, reports.
- **Not visible to users yet:** the Vercel and Render deployments have not been updated since 27 October, so none of the recent work is live.

2. Timeline

Date	Milestone
20 to 24 Sep	Both repositories started. Backend authentication: sign-up, sign-in, sign-out, email and OTP verification, password reset, multi-provider email, API documentation page.
28 to 30 Sep	User profiles and image uploads to AWS S3 (presigned URLs). Course category management with role-based access.
1 Oct	Course authoring and review pipeline, course enrolment and payment, payment webhooks, learning engine (progress, quiz grading, Q and A), real-time messaging and notifications, Express app setup and interactive API docs.
6 to 14 Oct	S3 public access fixed. Security hardening and real-time notifications. Phase 0 alignment (uploads, CORS, category images, token refresh, health checks). Trimester LMS API added (/api/lms).
14 to 15 Oct	Web LMS screens: weekly learning path, help centre, coding lab with in-browser editor and live code tab. Live classes streamed through YouTube. Student dashboard on live LMS data.
23 to 29 Oct	Free-course catalogue (74 courses) and public pages. Amazon SES email with redesigned Gloxad emails. Admin, tutor, institution onboarding and checkout screens. Deploy trigger on 29 Oct.
30 Oct	Certificate verification page, learner home, free-only course listing.
31 Oct to 2 Nov	Course player and quiz redesign to Figma, certificates, PWA install, responsive pass, new course cards. Trimester and Scholarships pages and backend (2 Nov).

WHAT HAS BEEN BUILT

3. Backend (gloxard-backend)

Express 5 with Mongoose and MongoDB. Roles: student, instructor, admin. Every private route checks the role and ownership on the server.

Area	What it provides
Authentication	Sign-up by role, sign-in and sign-out, email and OTP verification, forgot and reset password, token refresh, role guards. Email through Amazon SES with redesigned Gloxad templates.
Profiles and uploads	Profile update, instructor expertise, presigned uploads to AWS S3, signed URLs.
Courses	Categories (CRUD), course authoring, submit for review, admin approval and appeals, reviews and instructor replies, free, paid and trimester course types.
Enrolment and payment	Enrol, checkout and Paystack payment, webhook handling, transactions, instructor earnings, admin finance.
Learning engine	Progress per lesson, quiz submission and grading, notes, Q and A, announcements, certificate issue (needs 100 percent) and public certificate verification.
Messaging and notifications	Conversations and messages over sockets, notifications and mark-read, admin broadcast.
Trimester LMS (/api/lms)	Trimesters, cohorts, weekly plans, day items, progress, announcements, inquiries, live sessions (YouTube stream, chat, polls, hand raise, attendance, shared code), coding labs and submissions.
Scholarships and public pages (new)	Scholarship and application models, public trimester and scholarship lists, enrol, apply, admin create, edit, publish, archive and review, trimester status update.
Platform	Security hardening, CORS, rate limiters, health checks, interactive API documentation, seed scripts and a set of automated test scripts.

WHAT HAS BEEN BUILT

4. Web app (Gloxadapp)

Next.js 14 (App Router), TypeScript, Tailwind, React Query and Axios. Calls to /api are proxied to the backend.

Area	What it provides
Public site	Home, free courses (74), course pages, FAQs, tutor sign-up page, partner pages. Trimester Program and Scholarships landing pages (new).
Accounts	Sign-up by role, OTP verification, sign-in, forgot and reset password, role-aware header with a profile menu (Profile, Basic Information, Change Password, Log Out).
Course player	Left menu with modules and ticks, breadcrumb with previous and next, rounded video with YouTube controls, Full screen button, Mark as complete, resume. Tabs: Transcript, Notes, Course Materials, Review.
Quiz and results	All questions on one page, Submit enabled only when every question is answered, score ring, per-question results, Retake and Next Module.
Completion and certificates	Congratulations page, designed 800 by 600 certificate, PDF download, certificates list, public verify page that says whether a certificate is genuine.
Learner home and My Courses	Welcome banner, Continue Learning, Recommended, overall progress, In Progress, Completed and Certificates tabs. All numbers are real data.
Course cards	One shared clean card (Udemy and Coursera style) on the home page, Free Courses, Our Courses and the dashboard Free tab. Hover lift and heavy shadow removed.
Tutor, admin, institution	Tutor onboarding and course upload, admin dashboards (courses, students, tutors, finance, live, reviews, settings, notifications), institution onboarding flow, checkout and payment result pages.
Trimester LMS (/lms)	Role router to student, instructor or admin dashboard. Live data: student dashboard, weekly path, live class, coding lab, help centre. Sample data still: lesson, quiz, assignments, community, certificates, instructor and admin views.
PWA and responsiveness	Installable app (manifest, icons, offline page, service worker in production). Built screens checked at 375 px with no sideways scroll.
Course tiers	Trimester courses show as a programme with enrolment rules. 'Our Courses' stays hidden until a tutor has published a course; only free courses show until then.

THIS WEEK'S ASSIGNMENT

5. Trimester Program and LMS landing pages

Source: Gloxad LMS Weekly Developer Guide (2 November 2026). Day 1 audit is in docs/lms-audit.md.

Guide item	Status	Notes
Day 1: existing versus missing audit	Done	Written up; half of the LMS student screens are on live data, the rest on sample data.
Shared landing framework	Done	Header, loading, empty and error states, one shared date format.
Trimester landing page	Done	Published programmes only; four week journey; button by viewer state and server-checked window.
Scholarships landing page	Done	Published only; upcoming, open, closed; apply form; status with latest reviewer note.
Courses landing page	Existing	Course catalogue already built; not changed this week.
LMS home for visitors	Not started	/lms currently routes signed-in users to a dashboard.
Student lesson on real data	Not started	90 percent watched rule, sandbox autosave, assignment and discussion blocks.
Assignments and grading	Not started	No models yet: assignment, submission, rubric, grade, feedback.
Progress and Final Project	Not started	Completion ring, weights table, eight stage stepper.
Instructor and admin tools	Not started	Roster, attendance, grading, programme setup, scholarship review screens, reports.

Rules from the guide that are already honoured

- Permissions are enforced on the server; drafts are never public; visitors never see private progress.
- Submitted, reviewed, decided and completed stay separate; an application's history is appended, never overwritten.
- No dead buttons on the new pages; each call to action reflects the viewer's state.
- Every new page has loading, empty and error states and a clear success or error message.

QUALITY

6. What was tested

All tests below were run on a local copy (local database, seeded test users) on 2 November 2026.

Browser, desktop and phone

- Signed out: Trimester shows Sign in to enrol and four weeks; Scholarships shows Sign in to apply and an Opens on date.
- Student: Enrol now goes to the LMS dashboard; the page then shows Continue learning; applying for a scholarship shows Submitted.
- Earlier in the project: player, quiz and results end to end against the local API, completed page, real issued certificate, verify page, learner home, My Courses, free courses page.
- No horizontal overflow at 375 px on the player, quiz, certificate, completed, dashboard, My Courses, verify, Trimester and Scholarships pages.

Server rules

Check	Expected	Result
Enrol in the same trimester twice	409	Pass
Instructor tries to enrol	403	Pass
Signed-out user tries to enrol	401	Pass
Apply before the scholarship window opens	409	Pass
Apply after the window closes	409	Pass
Apply twice for the same scholarship	409	Pass
Student or tutor calls an admin route	403	Pass
Invalid scholarship id	400	Pass
Draft trimester or scholarship on the public page	Hidden	Pass
Admin changes an application status	History appended	Pass

Bugs found and fixed while testing

- Editing a scholarship without resending its lists wiped the eligibility and document lists.
- Scholarship dates showed one day early in some timezones; dates now display as calendar days.
- Earlier: a notes response shape that crashed the player; a card tag overlapping video titles.

Build checks: TypeScript is clean and the production build passes. The new work is not covered by automated tests in the repository yet.

WHAT IS NEXT

7. Outstanding work and decisions

Needed from the team

- **Trimester Dev Spec HTML** (the five student screens) and the 16 page PRD v1.0, so the lesson, live class and progress screens match the design.
- **Product decisions:** enrolment and payment (free or paid, prerequisites, withdrawal), completion rules, certificate issue and revocation rules, scholarship fields and review stages.
- **Designs** for instructor and admin screens, or agreement to reuse the student style.
- **GitHub:** the new organisation URL so the remote can be repointed and main pushed.
- **Hosting:** reconnect Vercel and Render to the repositories, redeploy, and run the catalogue import on Render. Open question: is the Vercel plan Hobby or Pro?

Backend follow-ups

- Return the correct answer index in the quiz submit breakdown so results can show the right answer.
- Add assignment, submission, rubric, grade and feedback models, a design-task upload endpoint, and program certificates.
- Add transcript and course-material fields, completion weights per course, and a server check of the 10 minute live class join window.
- Add AI Tutor, streak, weekly goal and discussions data (later phase; never a blocker).

Risks

- Nothing from the last week is deployed. All of it is local, and the course cards and this week's work are uncommitted in both repositories.
- A paid trimester refuses enrolment until checkout is connected.
- The seed password appears in a public repository and should be rotated or removed.
- The YouTube player terms of use still need technical lead review.

Proposed order for the next stage

Step	Work	Hand-over
1	Commit and deploy what exists; fix hosting	Live site shows current work
2	Lesson page, assignments, quiz on real data, Progress screen	A learner can finish a day
3	Instructor and admin tools, scholarship and programme screens	Staff can run a cohort
4	Integration, permissions, responsive and end-to-end QA	Demo and QA results